Professional Issues in IT

INVESTMENT APPRAISAL

Investment Appraisal

- Less resources and more number of proposals lead company owners and investors to decide which proposal to drop which to pick.
- The company will only have a limited amount of money of its own available and lenders and investors will only be prepared to offer limited amounts

Investing in.....

There is no single way of assessing and comparing the different proposals;

factors that must be taken into consideration include, for example:

- the extent to which the proposals are consistent with the company's long-term plans;
- the risk attached to the proposals;
- the availability of the necessary resources even if the money is available.

DCF(Discounted cash flow)

- One important criterion, however, is the financial one: which of the proposals will give the best return on the investment? The usual way of determining this is to use the method known as **discounted cash flow (DCF).**

DCF(Discounted cash flow)

It is important to realize that DCF is a tool that is used for many different purposes, for example:

- by investors on the stock market to assess whether the share price of a company reflects accurately its financial prospects;
- to assess whether it is better to purchase capital equipment or to lease it;
- to decide which of several possible projects is the most financially appealing;
- to decide whether a proposed capital project will be worthwhile.

THE TIME VALUE OF MONEY

- Money available at present time is worth more than the same amount in the future due to its potential earning capacity.
- Suppose that you have £100. You can choose to deposit it with a bank or some other savings organization. If the rate of interest is 3 per cent, then in a year's time you will have £103. In other words, the promise of £103 in a year's time is worth the same as £100 now.
- This simple example illustrates what is known as **the time value of money**. It forms the basis of discounted cash flow analysis.

Purchasing vs leasing

Suppose you want to get a car worth Rs.10,00,000/- and you have that amount.

The car finance department offers you to lease the car providing you pay a down payment of 20% that is Rs.2,00,000/- and a monthly payment of Rs.20,000 for 5 years(60 months).

Thus,

(20,000 * 60) + 2,00,000 = 14,00,000

While if you purchase it you have to pay Rs. 10,00,000/-

Would you purchase the car or lease it on these terms?

Interest rate

Suppose you put your Rs.10,00,000/- in the bank provided the bank gives you an interest of 10% per annum

Thus

(10/100) * 10,00,000 = 100,000 per annum

Discount factor

In general, if the interest rate is r (expressed as a fraction such as 0.08, not a percentage), then the present value of a sum of money X due in t years time is:

$$X \div (1 + r)^t$$

Or,

$$\frac{X}{(1+r)^t}$$

The quantity $1 \div (1 + r)^t$ is known as the *discount factor*.

TABLE 8.1 *Discount factors for periods up to five years*

Interest Rate (%)	1	2	3	4	5
3	0.9709	0.9426	0.9151	0.8885	0.8626
4	0.9615	0.9246	0.8890	0.8548	0.8219
5	0.9524	0.9070	0.8638	0.8227	0.7835
6	0.9434	0.8900	0.8396	0.7921	0.7473
7	0.9346	0.8734	0.8163	0.7629	0.7130
8	0.9259	0.8573	0.7938	0.7350	0.6806
9	0.9174	0.8417	0.7722	0.7084	0.6499
10	0.9091	0.8264	0.7513	0.6830	0.6209
15	0.8696	0.7561	0.6575	0.5718	0.4972
20	0.8333	0.6944	0.5787	0.4823	0.4019

Example

New Wolsey Hornet

£8995 or

only £500 down and £400 per month for 24 months

- The monthly rate is 0.2466 per cent per month.
- The net present value of £9,310.30 after calculating 24 payment of £ 400.
- To this we must add the £500 down payment. This shows that the NPV of the payments on easy terms is £9,810.30.

Example

A new van will cost £10,000. There will be annual costs of £500 for insurance and £150 for road tax. The cost of maintenance is estimated to be £200 in each of the first two years, £300 in year 3, £400 in year 4 and £500 in year 5.

At the end of the fifth year, it is expected that the van will be sold for around £2,000. The interest rate that the company pays on its borrowings is 10 percent.

If the van is hired(rented)

- Van hire costs £35 per day and it hires a van for about 100 days a year. All the costs are subject to inflation, which is judged to be around 5 per cent over the period, but the resale value of the van is the cash figure expected at the time.

TABLE 8.2 *DCF analysis of van purchase versus leasing*

	Year 0	Year 1	Year 2	Year 3	Year 4
Buying a van					
Van purchase/sale	(10000)				2000
Tax and insurance	(650)	(683)	(717)	(752)	(790)
Maintenance	(200)	(210)	(331)	(463)	(608)
Annual cash flow	(10850)	(893)	(1048)	(1215)	602
NPV of annual flow	(10850)	(812)	(866)	(914)	412
Total NPV	(13030)				
Continuing to rent					
Annual costs	(3500)	(3675)	(3859)	(4052)	(4254)
NPV of annual costs	(3500)	(3341)	(3189)	(3044)	(2906)
Total NPV	(15980)				

Cost of capital

Even if the company has the cash available to buy the van outright, there is still a cost because the company will lose the income it could have received by investing the money somewhere else, in a suitable interest bearing account for example. Such a cost is known as an *opportunity cost*. *If the company* is able to pay cash for the van, this is the interest rate it would be appropriate to use in the DCF analysis.

Handling inflation

What is inflation?

Inflation in a financial context means the fall in the value of money over time

The presence of inflation means that the ‘monetary’ rate of interest, that is, the rate that is normally quoted is something of a delusion. £100 invested at a quoted interest rate of 10 per cent will be worth £110 in money in a years’ time.

However, if the rate of inflation is 5 per cent this £110 will only buy as much as £110/1.05 = £104.76 would buy today. Thus the real rate of interest is only 4.76 per cent.

Software company example

We consider a company that is assessing a proposal for the development of a software product. It is estimated that three people will be required for development in the first year and a further person and a half in the second year; suitable staff cost £35,000 per year, including the employer's pension and national insurance costs. The product will be released in the second year. After the second year, maintenance is expected to require one person, full-time. Sales and marketing costs are estimated to be £10,000 in the first year, rising to £20,000 for each of the next four years.

The product itself is a fairly high-value but specialized product. It is expected that about 100 copies will be sold over this period, at around £5,000 a copy.

TABLE 8.3 *DCF analysis of a proposed software package development*

	Year 0	Year 1	Year 2	Year 3	Year 4
Development cost	105000	55125			
Maintenance			38588	40517	42543
Sales and marketing	10000	21000	22050	23153	24310
Number of sales		10	20	40	30
Revenue		50000	100000	200000	150000
Net cash flow	(115000)	(26125)	39363	136331	83147
Discount factor	1	0.9091	0.8264	0.7513	0.6830
Present value	(115000)	(23750)	32529	102425	56789
Cumulative present value	(115000)	(138750)	(106221)	(3796)	52993

PITFALLS OF DCF

we must take into account that:

most software projects take more effort than expected;

most software doesn’t work very well when it’s first released;

we may not manage to sell as many copies as we expected;

there is a considerable risk that a competitor will launch a similar product before ours is ready.